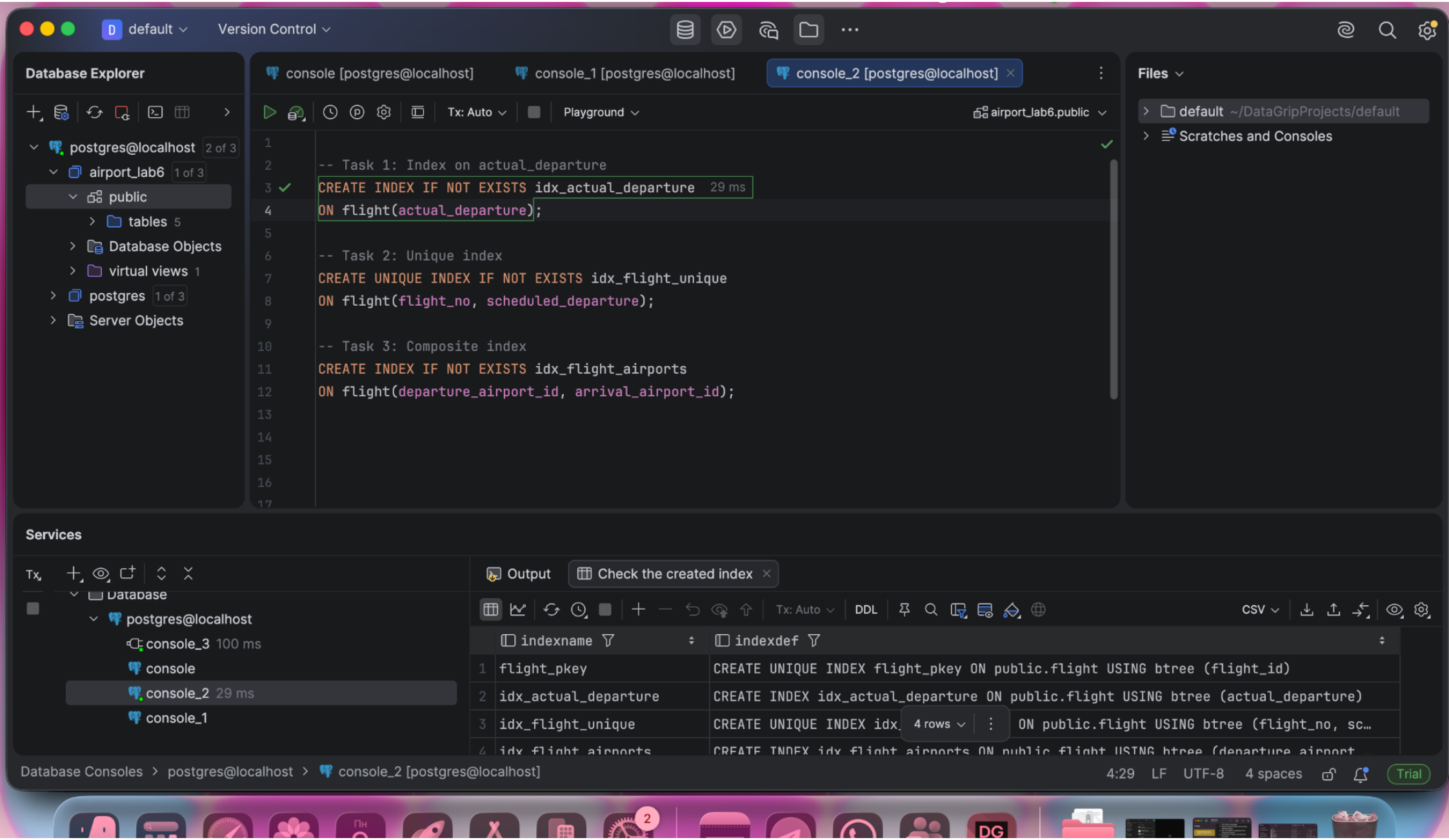
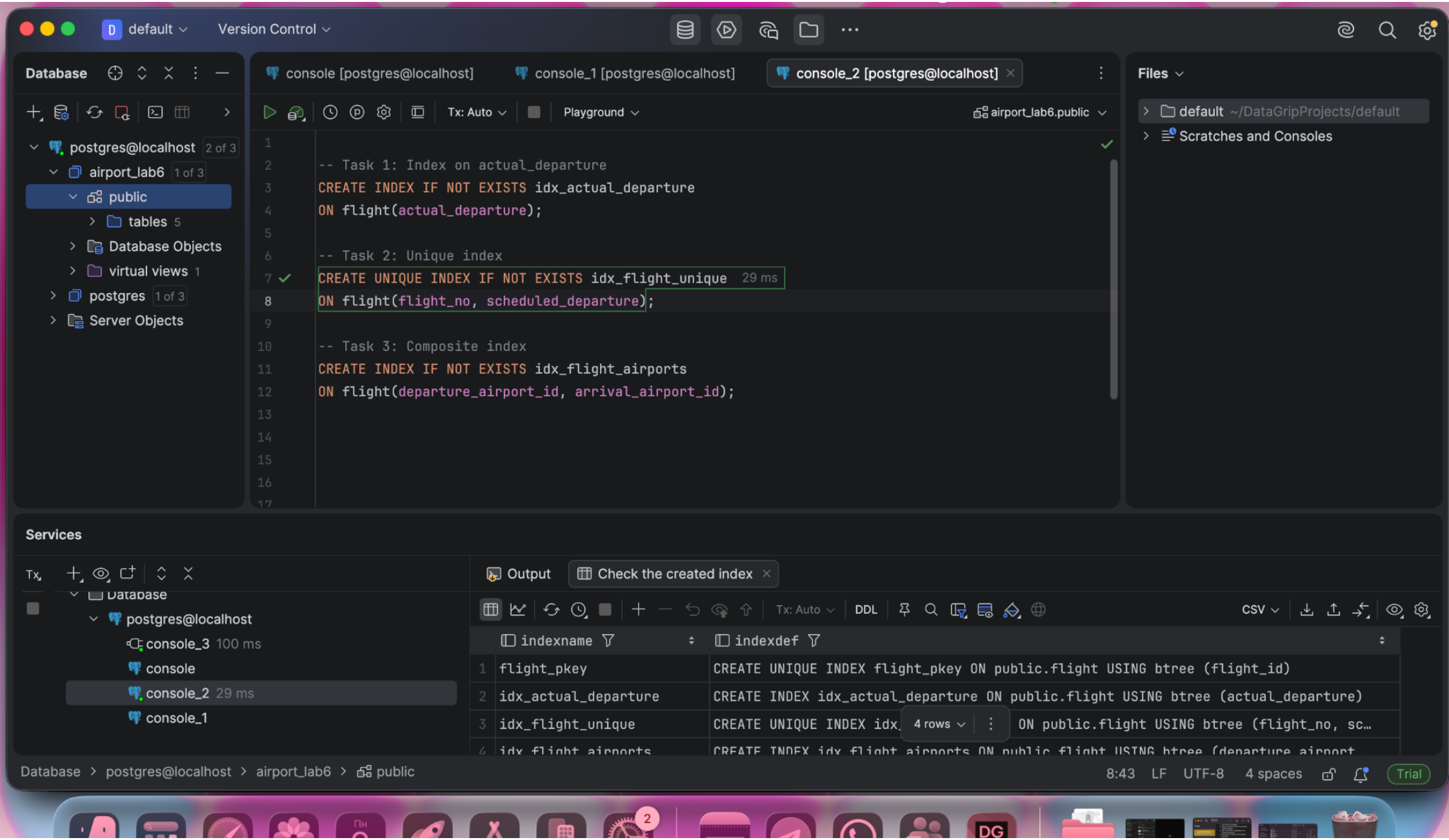
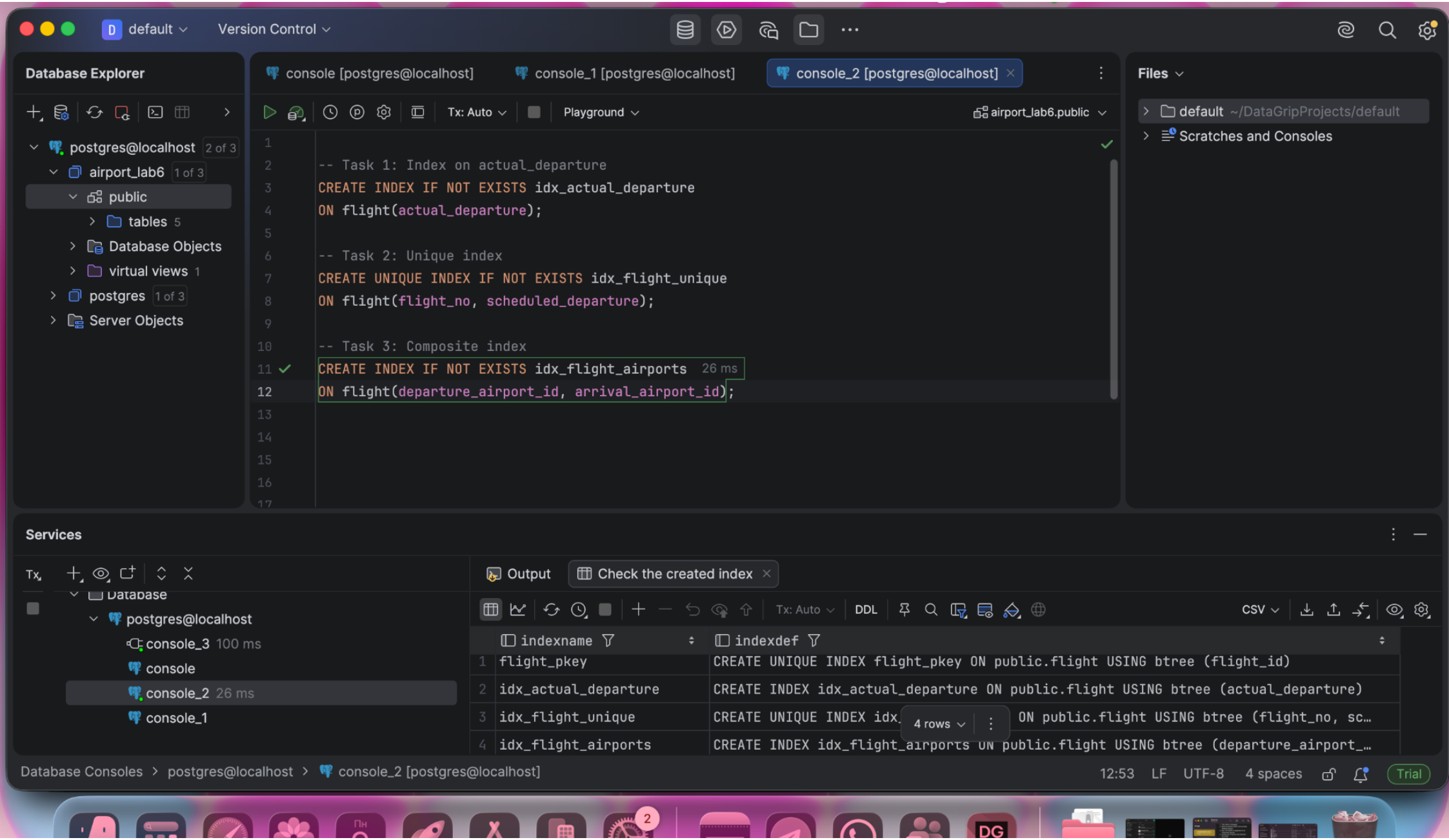
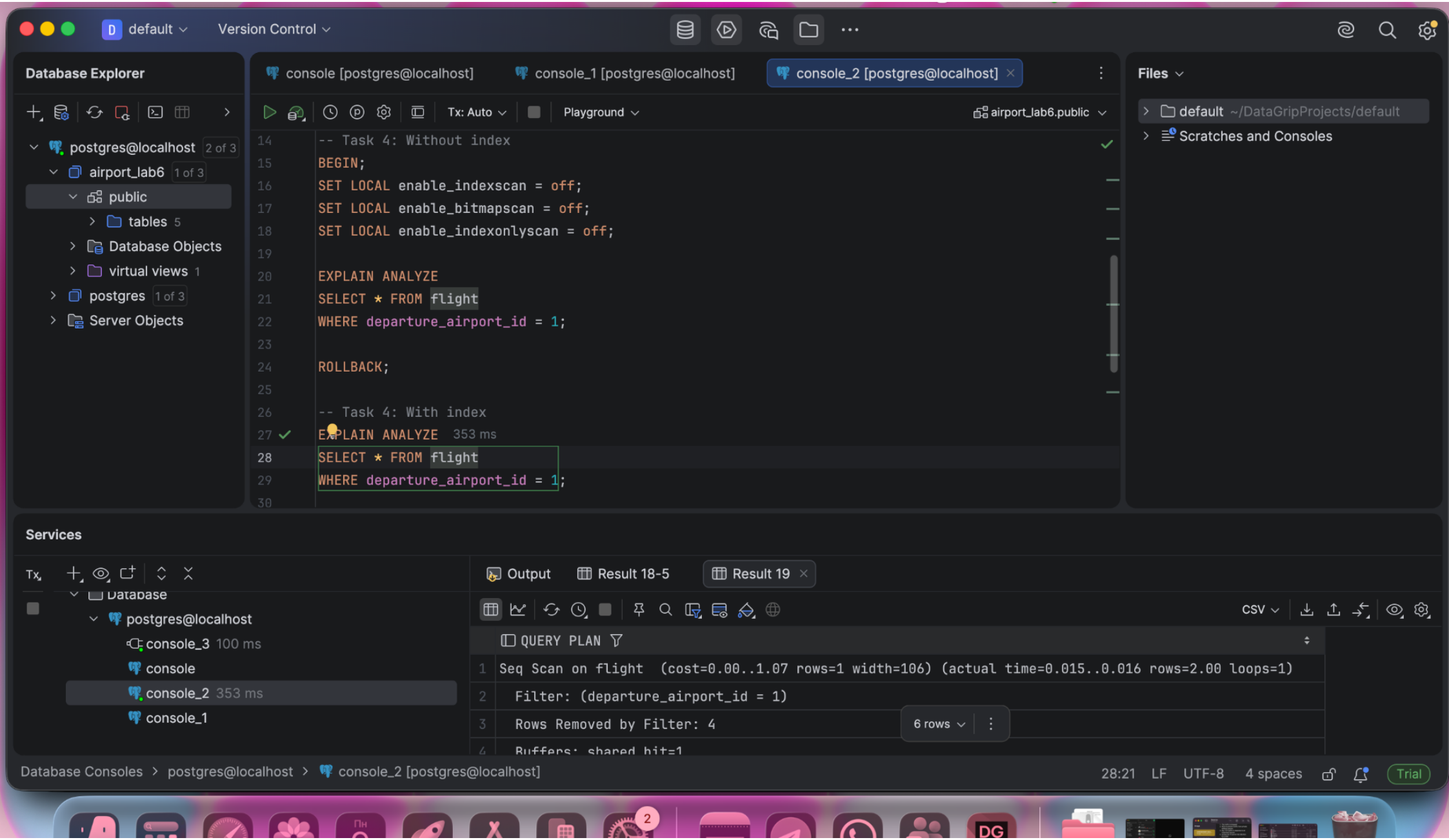
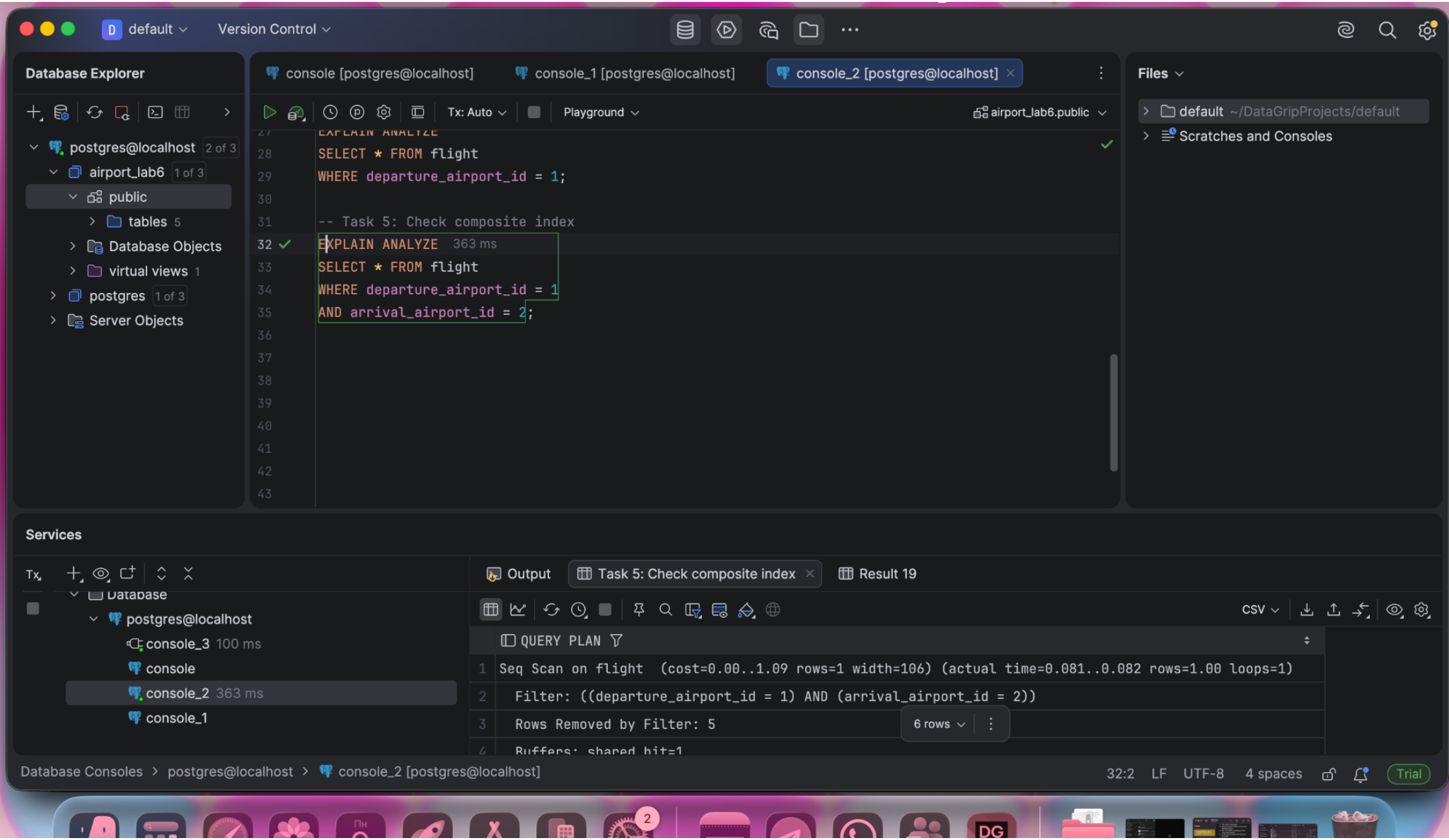
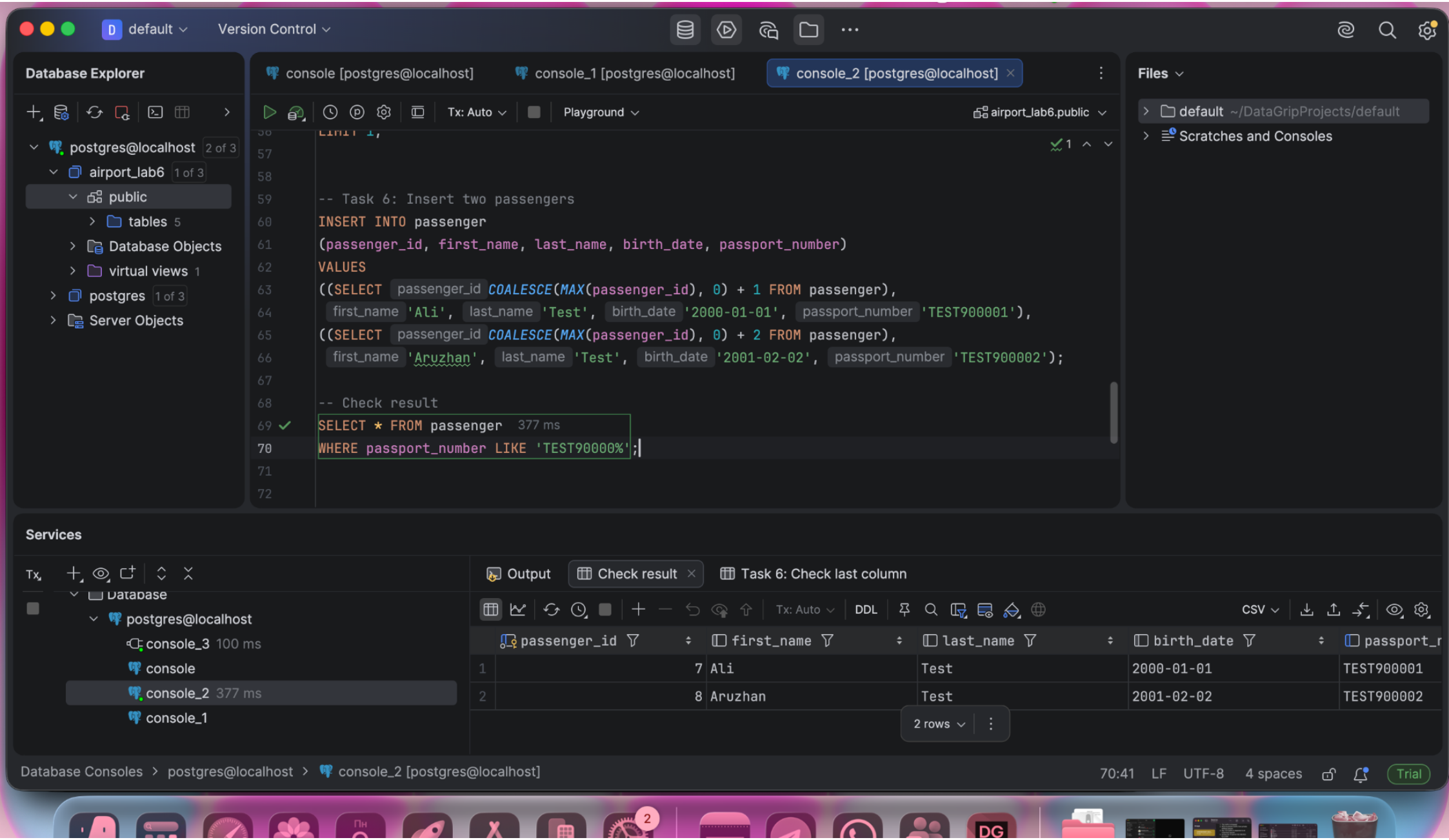

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console [postgres@localhost]
console_1 [postgres@localhost]
console_2 [postgres@localhost]
Tx: Auto
Playground
airport_lab6.public
-- Task 1: Index on actual_departure
CREATE INDEX IF NOT EXISTS idx_actual_departure 29 ms
ON flight(actual_departure);
-- Task 2: Unique index
CREATE UNIQUE INDEX IF NOT EXISTS idx_flight_unique
ON flight(flight_no, scheduled_departure);
-- Task 3: Composite index
CREATE INDEX IF NOT EXISTS idx_flight_airports
ON flight(departure_airport_id, arrival_airport_id);
Files
default ~/DataGripProjects/default
Scratches and Consoles
Services
postgres@localhost
console_3 100 ms
console
console_2 29 ms
console_1
Output
Check the created index
Tx: Auto
DDL
CSV
indexname
indexdef
1 flight_pkey CREATE UNIQUE INDEX flight_pkey ON public.flight USING btree (flight_id)
2 idx_actual_departure CREATE INDEX idx_actual_departure ON public.flight USING btree (actual_departure)
3 idx_flight_unique CREATE UNIQUE INDEX idx
4 rows
ON public.flight USING btree (flight_no, sc…
Database Consoles > postgres@localhost > console_2 [postgres@localhost]
4:29 LF UTF-8 4 spaces
Trial

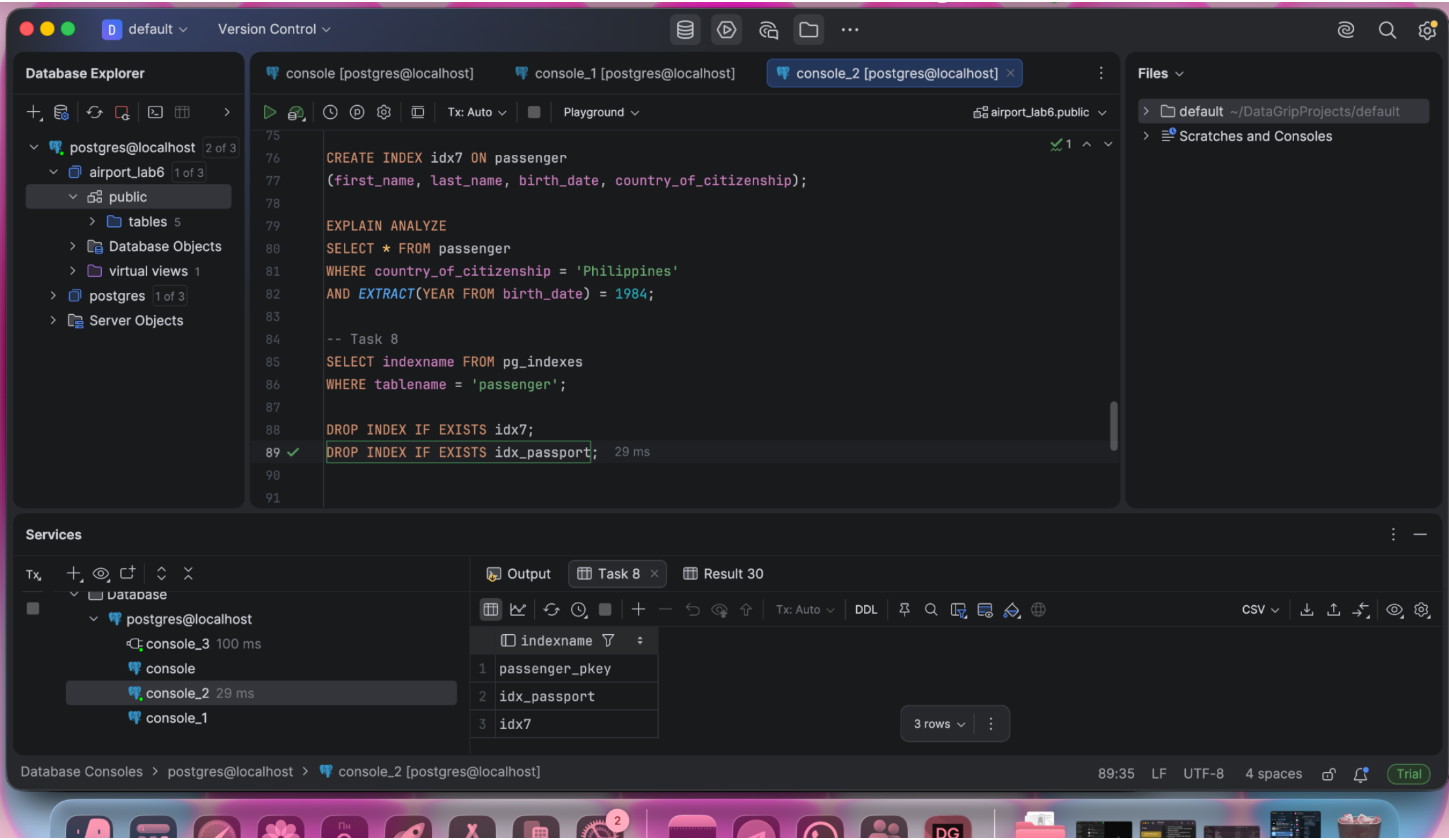

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console_1 [postgres@localhost]
console_2 [postgres@localhost]
Tx: Auto
Playground
airport_lab6.public
-- Task 1: Index on actual_departure
CREATE INDEX IF NOT EXISTS idx_actual_departure
ON flight(actual_departure);
-- Task 2: Unique index
CREATE UNIQUE INDEX IF NOT EXISTS idx_flight_unique 29 ms
ON flight(flight_no, scheduled_departure);
-- Task 3: Composite index
CREATE INDEX IF NOT EXISTS idx_flight_airports
ON flight(departure_airport_id, arrival_airport_id);
Files
default ~/DataGripProjects/default
Scratches and Consoles
Services
postgres@localhost
console_3 100 ms
console
console_2 29 ms
console_1
Output
Check the created index
Tx: Auto
DDL
CSV
indexname
indexdef
1 flight_pkey CREATE UNIQUE INDEX flight_pkey ON public.flight USING btree (flight_id)
2 idx_actual_departure CREATE INDEX idx_actual_departure ON public.flight USING btree (actual_departure)
3 idx_flight_unique CREATE UNIQUE INDEX idx
4 rows
ON public.flight USING btree (flight_no, sc…
Database > postgres@localhost > airport_lab6 > public
8:43 LF UTF-8 4 spaces
Trial

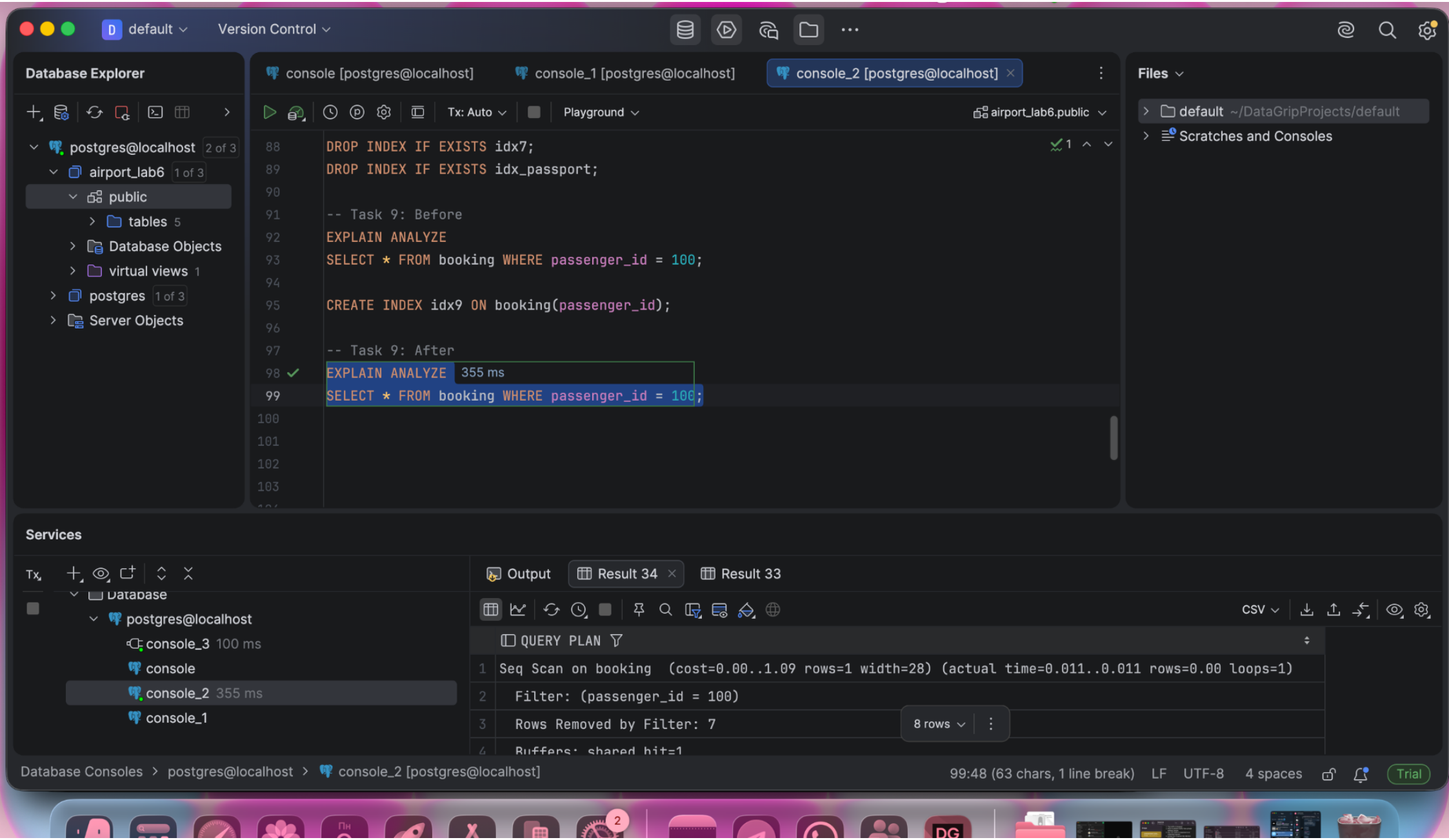

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Server Objects
console [postgres@localhost]
console_1 [postgres@localhost]
console_2 [postgres@localhost]
Tx: Auto
Playground
airport_lab6.public
-- Task 1: Index on actual_departure
CREATE INDEX IF NOT EXISTS idx_actual_departure
ON flight(actual_departure);
-- Task 2: Unique index
CREATE UNIQUE INDEX IF NOT EXISTS idx_flight_unique
ON flight(flight_no, scheduled_departure);
-- Task 3: Composite index
CREATE INDEX IF NOT EXISTS idx_flight_airports 26 ms
ON flight(departure_airport_id, arrival_airport_id);
Files
default ~/DataGripProjects/default
Scratches and Consoles
Services
Database
postgres@localhost
console_3 100 ms
console
console_2 26 ms
console_1
Output
Check the created index
Tx: Auto
DDL
CSV
indexname
indexdef
1 flight_pkey CREATE UNIQUE INDEX flight_pkey ON public.flight USING btree (flight_id)
2 idx_actual_departure CREATE INDEX idx_actual_departure ON public.flight USING btree (actual_departure)
3 idx_flight_unique CREATE UNIQUE INDEX idx… ON public.flight USING btree (flight_no, sc…
4 idx_flight_airports CREATE INDEX idx_flight_airports ON public.flight USING btree (departure_airport_…
4 rows
Database Consoles > postgres@localhost > console_2 [postgres@localhost]
12:53 LF UTF-8 4 spaces
Trial

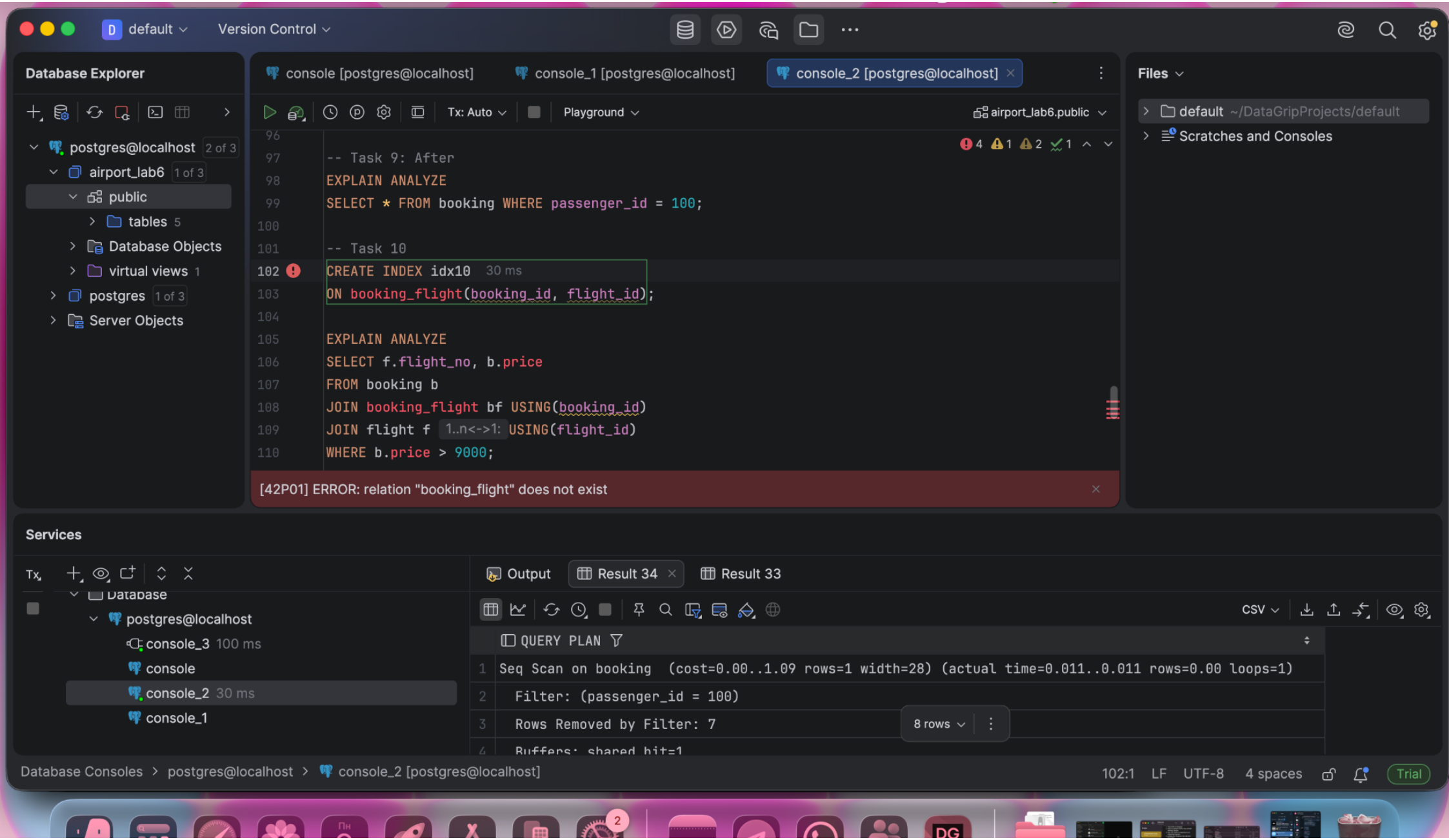

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console [postgres@localhost]
console_1 [postgres@localhost]
console_2 [postgres@localhost]
Tx: Auto
Playground
airport_lab6.public
-- Task 4: Without index
BEGIN;
SET LOCAL enable_indexscan = off;
SET LOCAL enable_bitmapscan = off;
SET LOCAL enable_indexonlyscan = off;
EXPLAIN ANALYZE
SELECT * FROM flight
WHERE departure_airport_id = 1;
ROLLBACK;
-- Task 4: With index
EXPLAIN ANALYZE 353 ms
SELECT * FROM flight
WHERE departure_airport_id = 1;
Files
default ~/DataGripProjects/default
Scratches and Consoles
Services
Database
postgres@localhost
console_3 100 ms
console
console_2 353 ms
console_1
Output
Result 18-5
Result 19
CSV
QUERY PLAN
1 Seq Scan on flight (cost=0.00..1.07 rows=1 width=106) (actual time=0.015..0.016 rows=2.00 loops=1)
2 Filter: (departure_airport_id = 1)
3 Rows Removed by Filter: 4
6 rows
Database Consoles > postgres@localhost > console_2 [postgres@localhost]
28:21 LF UTF-8 4 spaces
Trial

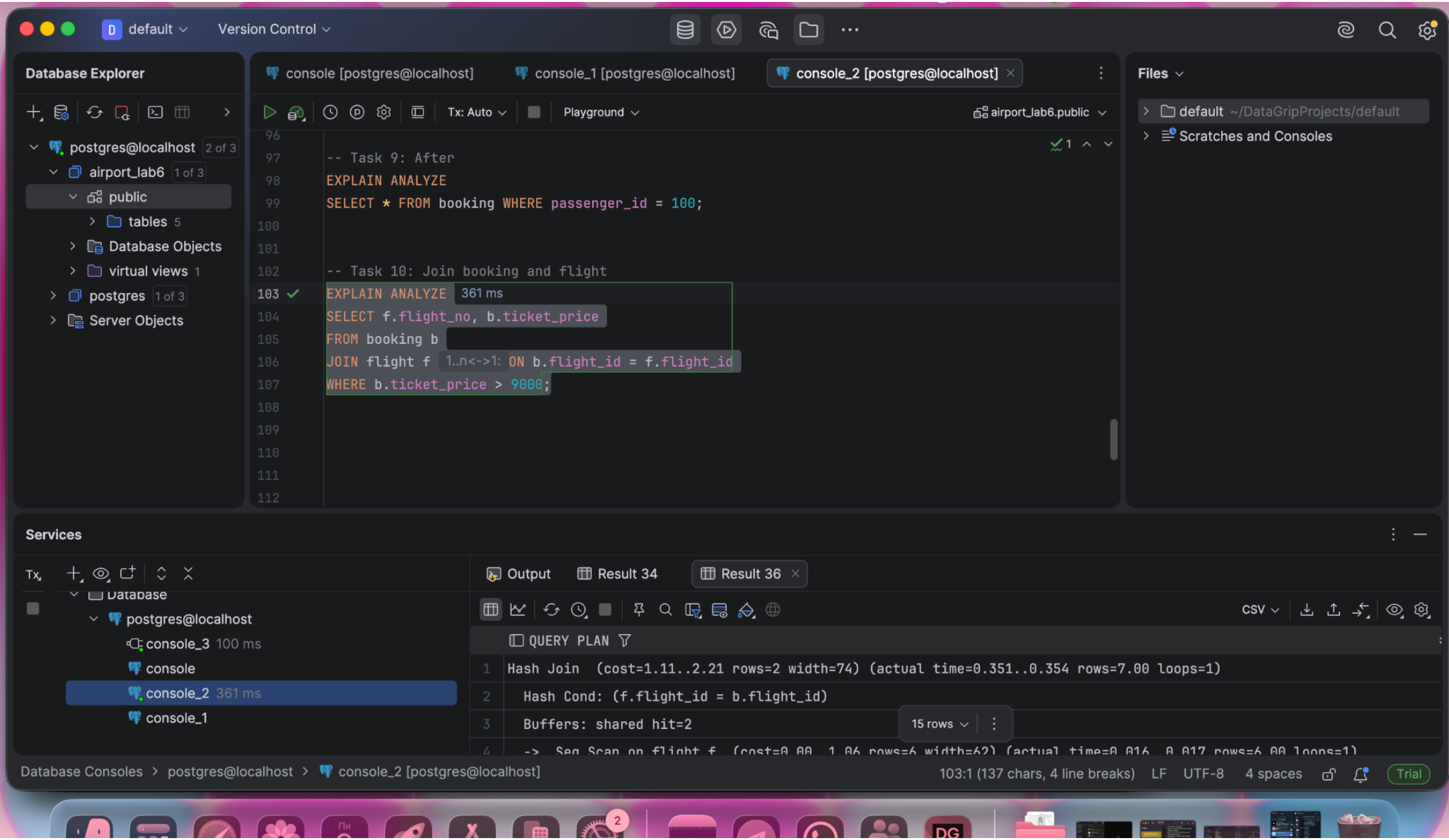

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Server Objects
console [postgres@localhost]
console_1 [postgres@localhost]
console_2 [postgres@localhost]
Tx: Auto
Playground
airport_lab6.public
28 SELECT * FROM flight
29 WHERE departure_airport_id = 1;
31 -- Task 5: Check composite index
32 EXPLAIN ANALYZE 363 ms
33 SELECT * FROM flight
34 WHERE departure_airport_id = 1
35 AND arrival_airport_id = 2;
Files
default ~/DataGripProjects/default
Scratches and Consoles
Services
Database
postgres@localhost
console_3 100 ms
console
console_2 363 ms
console_1
Output
Task 5: Check composite index
Result 19
CSV
QUERY PLAN
1 Seq Scan on flight (cost=0.00..1.09 rows=1 width=106) (actual time=0.081..0.082 rows=1.00 loops=1)
2 Filter: ((departure_airport_id = 1) AND (arrival_airport_id = 2))
3 Rows Removed by Filter: 5
6 rows
Database Consoles > postgres@localhost > console_2 [postgres@localhost]
32:2 LF UTF-8 4 spaces
Trial

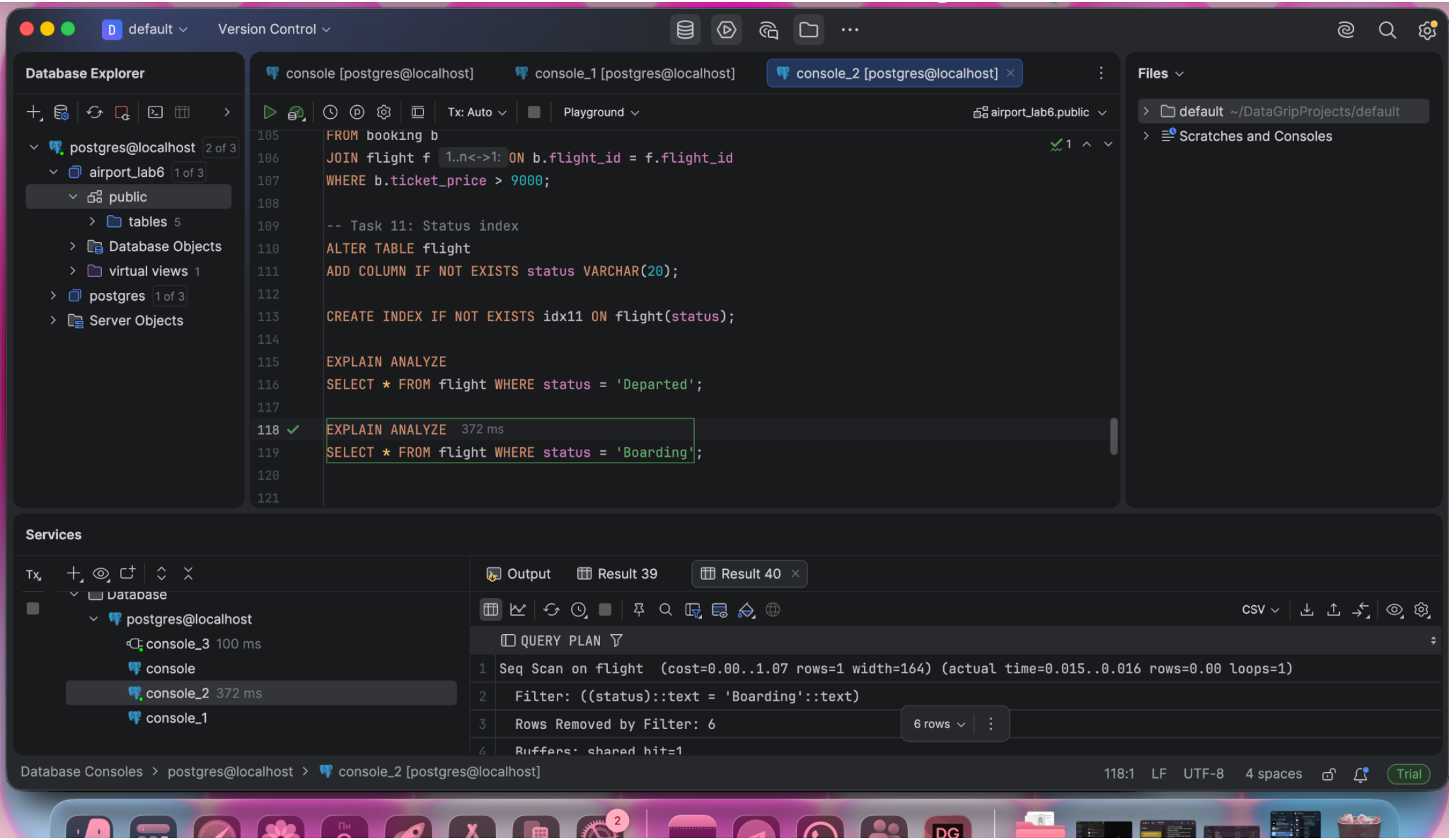

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console [postgres@localhost]
console_1 [postgres@localhost]
console_2 [postgres@localhost]
Tx: Auto
Playground
airport_lab6.public
LIMIT 1;
-- Task 6: Insert two passengers
INSERT INTO passenger
(passenger_id, first_name, last_name, birth_date, passport_number)
VALUES
((SELECT passenger_id COALESCE(MAX(passenger_id), 0) + 1 FROM passenger),
first_name 'Ali', last_name 'Test', birth_date '2000-01-01', passport_number 'TEST900001'),
((SELECT passenger_id COALESCE(MAX(passenger_id), 0) + 2 FROM passenger),
first_name 'Aruzhan', last_name 'Test', birth_date '2001-02-02', passport_number 'TEST900002');
-- Check result
SELECT * FROM passenger 377 ms
WHERE passport_number LIKE 'TEST90000%';
Files
default ~/DataGripProjects/default
Scratches and Consoles
Services
Database
postgres@localhost
console_3 100 ms
console
console_2 377 ms
console_1
Output
Check result
Task 6: Check last column
Tx: Auto
DDL
CSV
passenger_id
first_name
last_name
birth_date
passport_r
1 7 Ali Test 2000-01-01 TEST900001
2 8 Aruzhan Test 2001-02-02 TEST900002
2 rows
Database Consoles > postgres@localhost > console_2 [postgres@localhost]
70:41 LF UTF-8 4 spaces
Trial

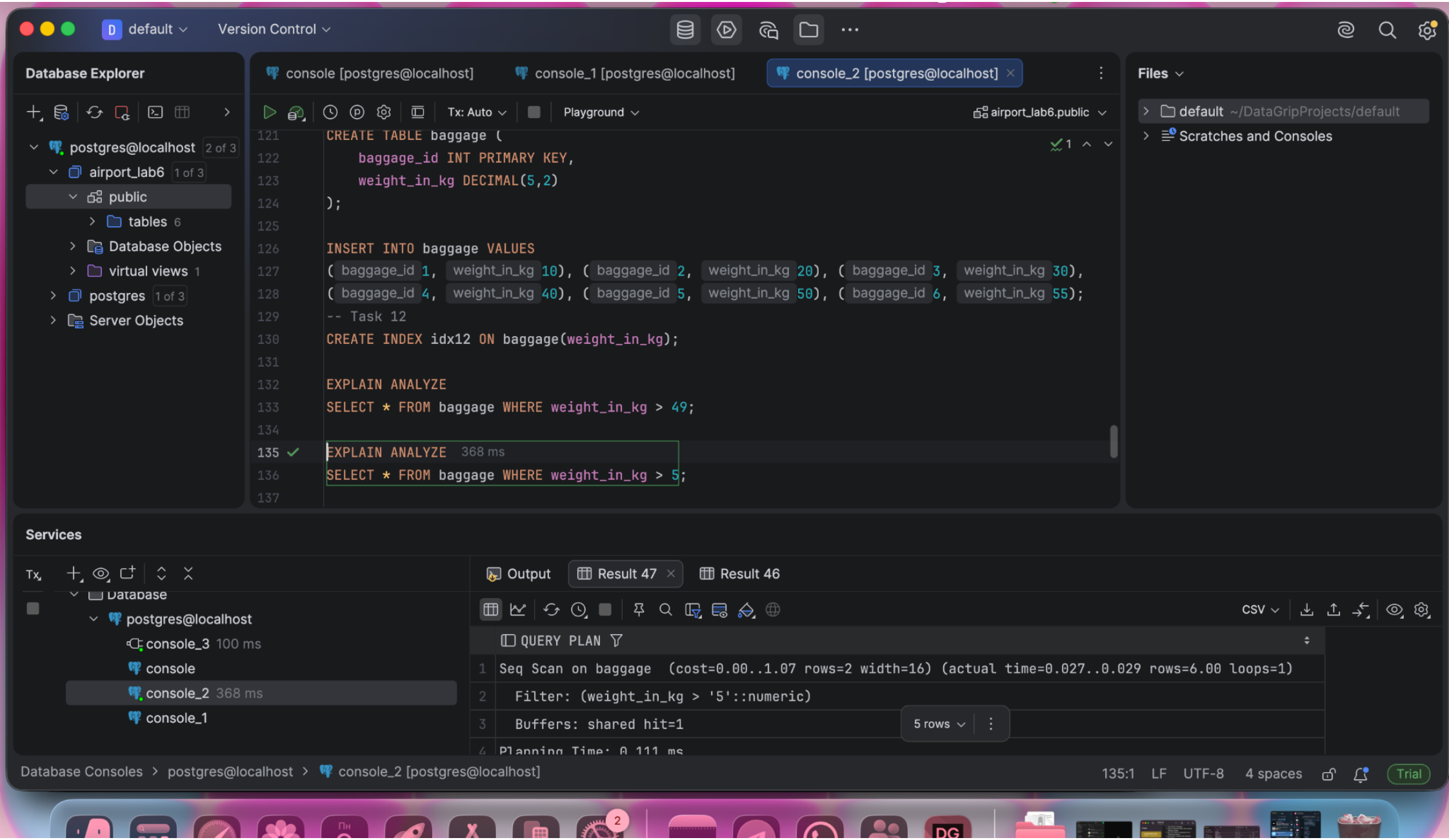

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console_2 [postgres@localhost]
Tx: Auto
Playground
airport_lab6.public
CREATE INDEX idx7 ON passenger
(first_name, last_name, birth_date, country_of_citizenship);
EXPLAIN ANALYZE
SELECT * FROM passenger
WHERE country_of_citizenship = 'Philippines'
AND EXTRACT(YEAR FROM birth_date) = 1984;
-- Task 8
SELECT indexname FROM pg_indexes
WHERE tablename = 'passenger';
DROP INDEX IF EXISTS idx7;
DROP INDEX IF EXISTS idx_passport; 29 ms
Files
default ~/DataGripProjects/default
Scratches and Consoles
Services
Database
postgres@localhost
console_3 100 ms
console
console_2 29 ms
console_1
Output
Task 8
Result 30
Tx: Auto
DDL
CSV
indexname
1 passenger_pkey
2 idx_passport
3 idx7
3 rows
Database Consoles > postgres@localhost > console_2 [postgres@localhost]
89:35 LF UTF-8 4 spaces
Trial

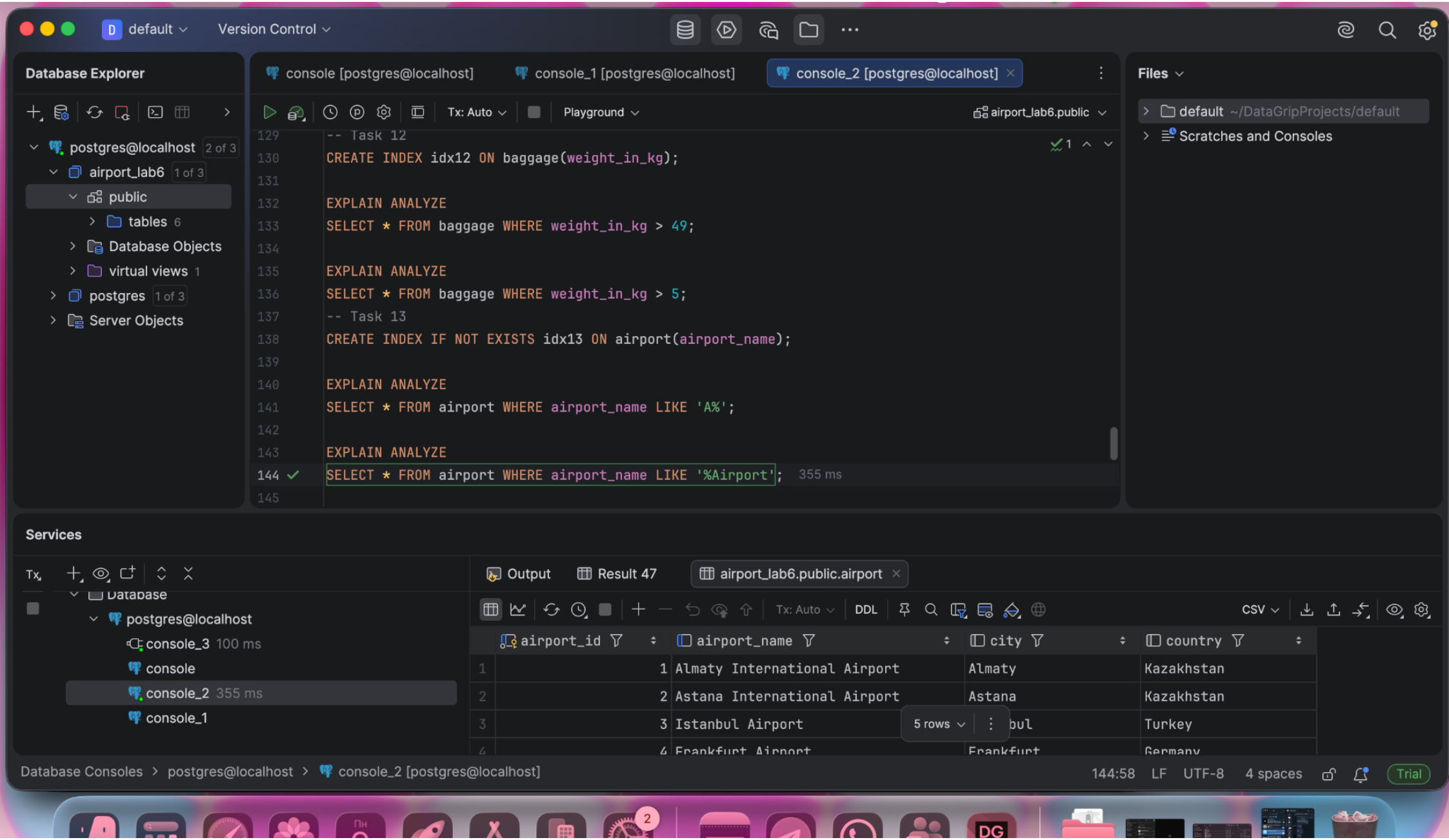

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console [postgres@localhost]
console_1 [postgres@localhost]
console_2 [postgres@localhost]
Tx: Auto
Playground
airport_lab6.public
88 DROP INDEX IF EXISTS idx7;
89 DROP INDEX IF EXISTS idx_passport;
90
91 -- Task 9: Before
92 EXPLAIN ANALYZE
93 SELECT * FROM booking WHERE passenger_id = 100;
94
95 CREATE INDEX idx9 ON booking(passenger_id);
96
97 -- Task 9: After
98 EXPLAIN ANALYZE 355 ms
99 SELECT * FROM booking WHERE passenger_id = 100;
100
101
102
103
Files
default ~/DataGripProjects/default
Scratches and Consoles
Services
Database
postgres@localhost
console_3 100 ms
console
console_2 355 ms
console_1
Output
Result 34
Result 33
CSV
QUERY PLAN
1 Seq Scan on booking (cost=0.00..1.09 rows=1 width=28) (actual time=0.011..0.011 rows=0.00 loops=1)
2 Filter: (passenger_id = 100)
3 Rows Removed by Filter: 7
8 rows
Database Consoles > postgres@localhost > console_2 [postgres@localhost]
99:48 (63 chars, 1 line break)
LF
UTF-8
4 spaces
Trial

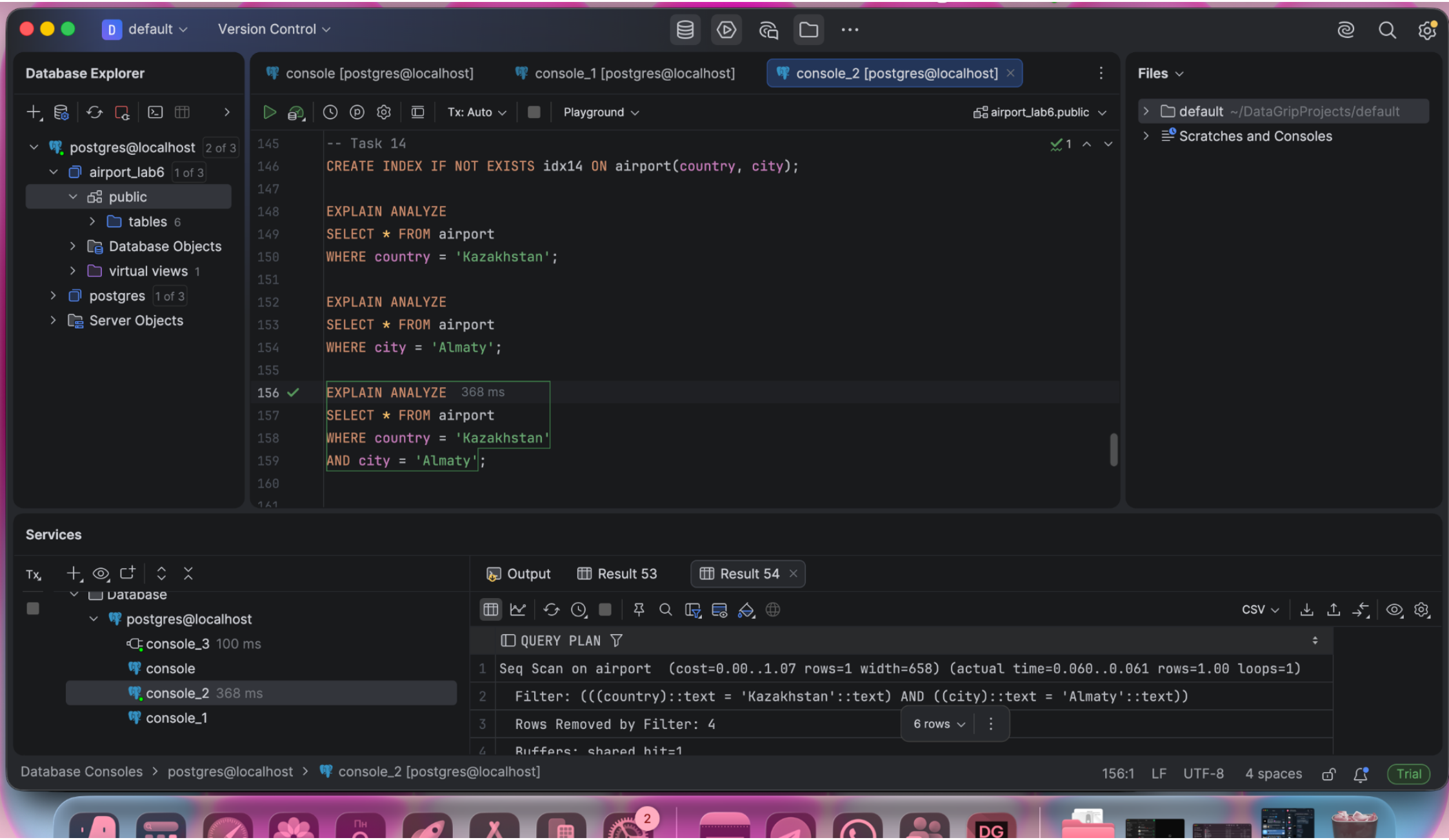

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console [postgres@localhost]
console_1 [postgres@localhost]
console_2 [postgres@localhost]
Tx: Auto
Playground
airport_lab6.public
-- Task 9: After
EXPLAIN ANALYZE
SELECT * FROM booking WHERE passenger_id = 100;
-- Task 10
CREATE INDEX idx10 30 ms
ON booking_flight(booking_id, flight_id);
EXPLAIN ANALYZE
SELECT f.flight_no, b.price
FROM booking b
JOIN booking_flight bf USING(booking_id)
JOIN flight f 1..n<->1: USING(flight_id)
WHERE b.price > 9000;
[42P01] ERROR: relation "booking_flight" does not exist
Files
default ~/DataGripProjects/default
Scratches and Consoles
Services
Database
postgres@localhost
console_3 100 ms
console
console_2 30 ms
console_1
Output
Result 34
Result 33
CSV
QUERY PLAN
Seq Scan on booking (cost=0.00..1.09 rows=1 width=28) (actual time=0.011..0.011 rows=0.00 loops=1)
Filter: (passenger_id = 100)
Rows Removed by Filter: 7
8 rows
Database Consoles > postgres@localhost > console_2 [postgres@localhost]
102:1 LF UTF-8 4 spaces
Trial

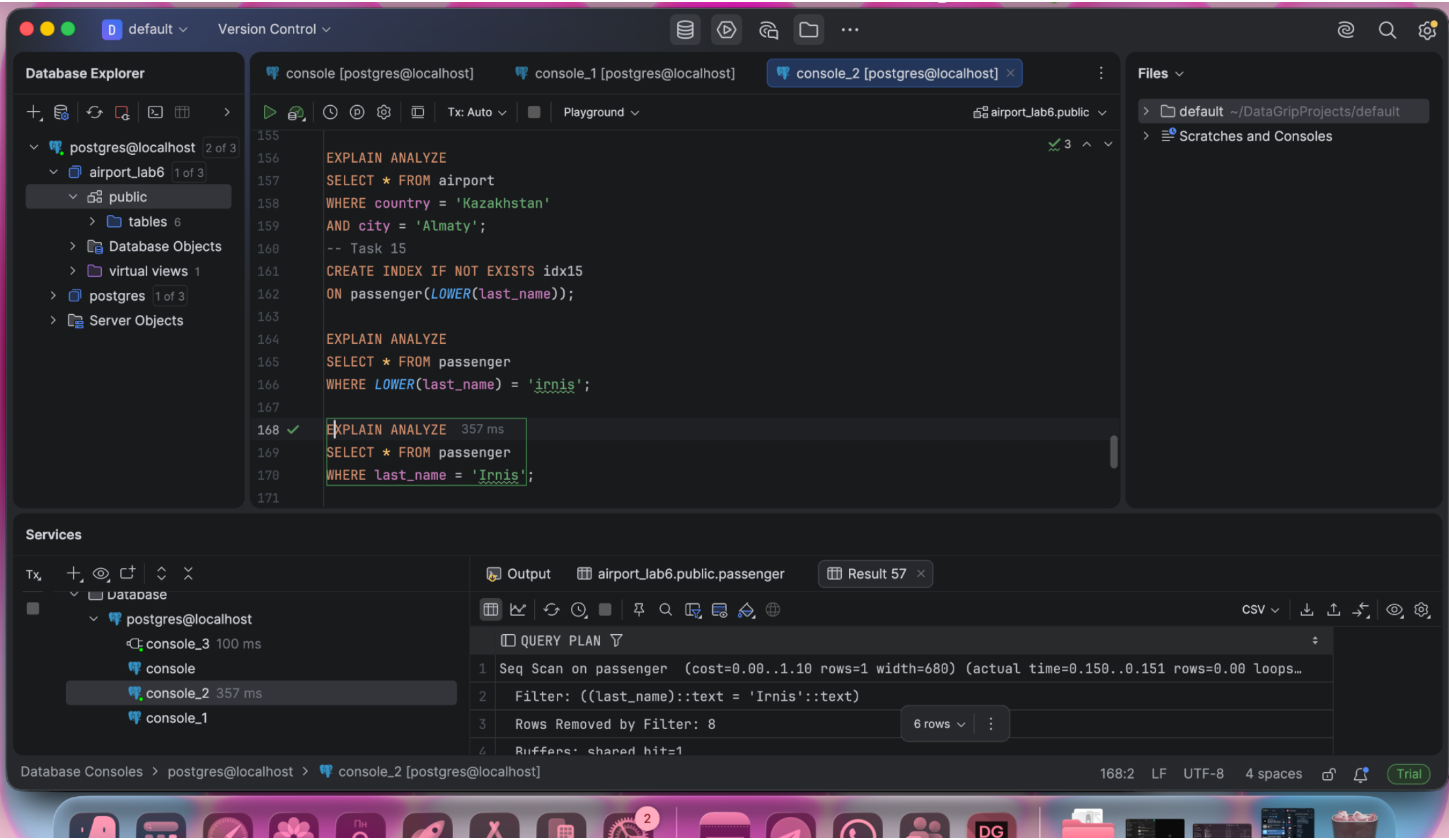

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console_1 [postgres@localhost]
console_2 [postgres@localhost]
Tx: Auto
Playground
airport_lab6.public
-- Task 9: After
EXPLAIN ANALYZE
SELECT * FROM booking WHERE passenger_id = 100;
-- Task 10: Join booking and flight
EXPLAIN ANALYZE 361 ms
SELECT f.flight_no, b.ticket_price
FROM booking b
JOIN flight f 1..n<->1: ON b.flight_id = f.flight_id
WHERE b.ticket_price > 9000;
Files
default ~/DataGripProjects/default
Scratches and Consoles
Services
Database
postgres@localhost
console_3 100 ms
console
console_2 361 ms
console_1
Output
Result 34
Result 36
CSV
QUERY PLAN
Hash Join (cost=1.11..2.21 rows=2 width=74) (actual time=0.351..0.354 rows=7.00 loops=1)
Hash Cond: (f.flight_id = b.flight_id)
Buffers: shared hit=2
15 rows
Database Consoles > postgres@localhost > console_2 [postgres@localhost]
103:1 (137 chars, 4 line breaks) LF UTF-8 4 spaces
Trial

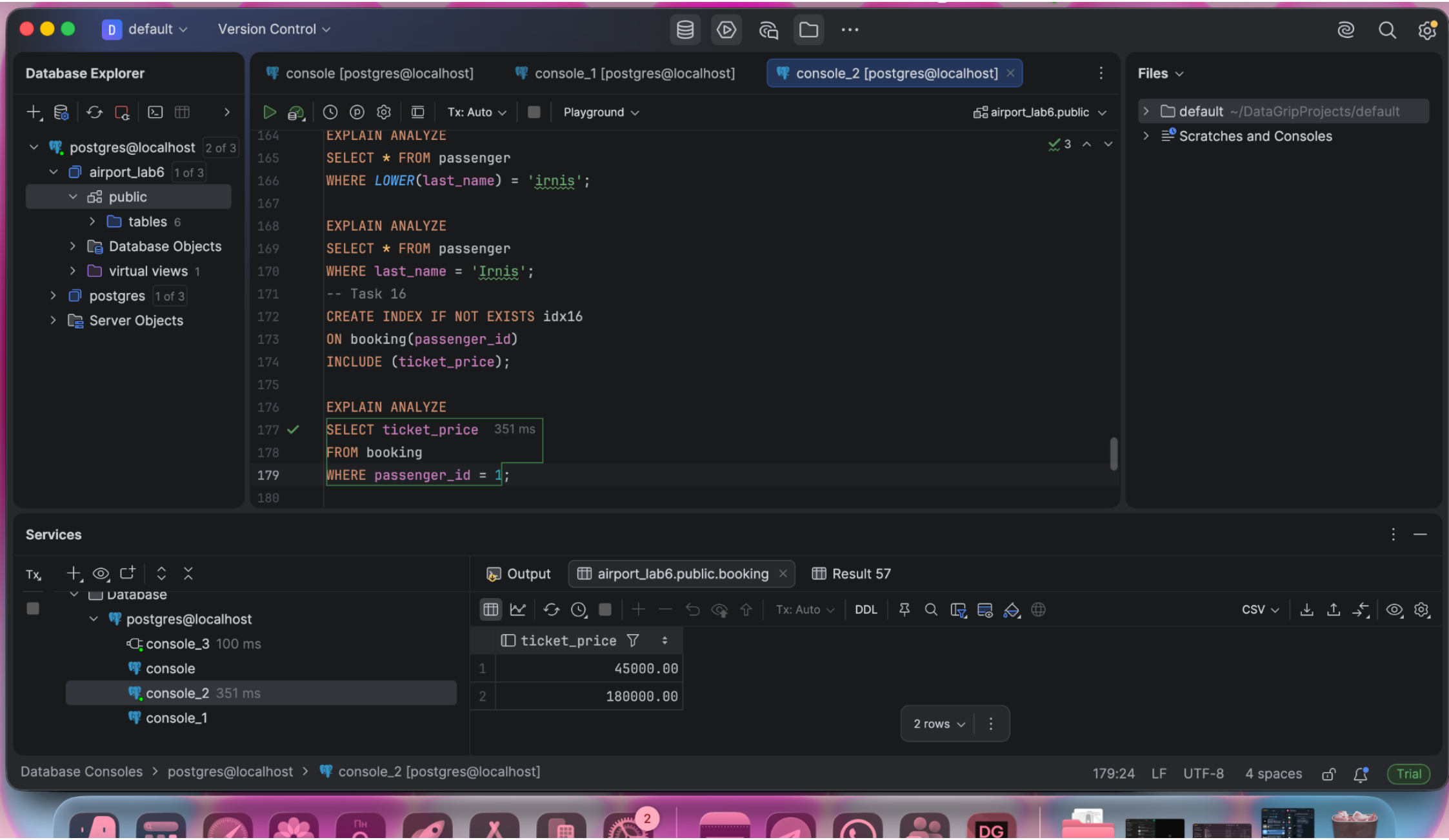

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console_2 [postgres@localhost]
Tx: Auto
Playground
airport_lab6.public
FROM booking b
JOIN flight f 1..n<->1: ON b.flight_id = f.flight_id
WHERE b.ticket_price > 9000;
-- Task 11: Status index
ALTER TABLE flight
ADD COLUMN IF NOT EXISTS status VARCHAR(20);
CREATE INDEX IF NOT EXISTS idx11 ON flight(status);
EXPLAIN ANALYZE
SELECT * FROM flight WHERE status = 'Departed';
EXPLAIN ANALYZE 372 ms
SELECT * FROM flight WHERE status = 'Boarding';
Files
default ~/DataGripProjects/default
Scratches and Consoles
Services
Database
postgres@localhost
console_3 100 ms
console
console_2 372 ms
console_1
Output
Result 39
Result 40
CSV
QUERY PLAN
Seq Scan on flight (cost=0.00..1.07 rows=1 width=164) (actual time=0.015..0.016 rows=0.00 loops=1)
Filter: ((status)::text = 'Boarding'::text)
Rows Removed by Filter: 6
6 rows
Buffers: shared hit=1
Database Consoles > postgres@localhost > console_2 [postgres@localhost]
118:1 LF UTF-8 4 spaces
Trial

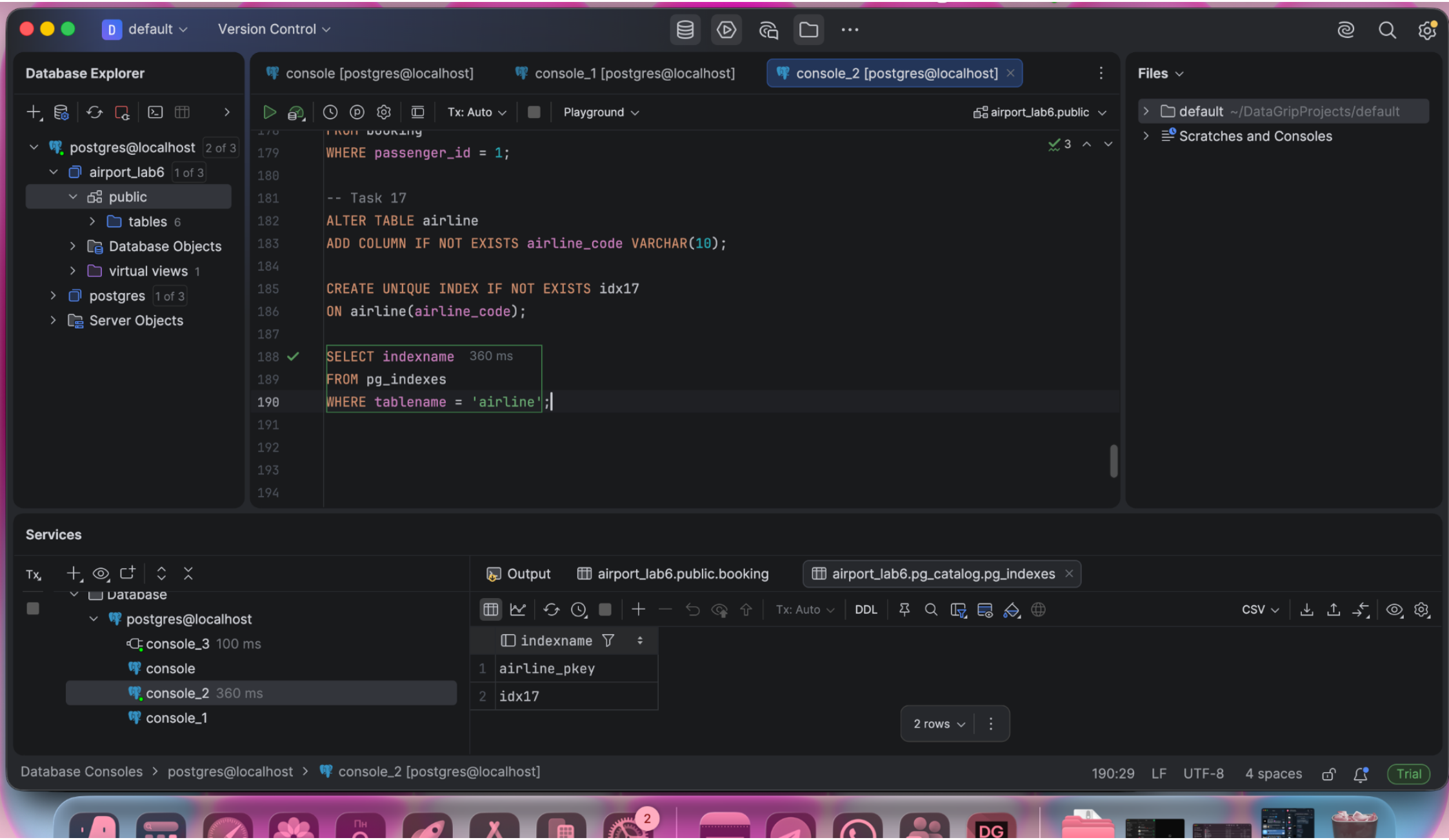

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Server Objects
console [postgres@localhost]
console_1 [postgres@localhost]
console_2 [postgres@localhost]
Tx: Auto
Playground
airport_lab6.public
CREATE TABLE baggage (
baggage_id INT PRIMARY KEY,
weight_in_kg DECIMAL(5,2)
);
INSERT INTO baggage VALUES
(baggage_id 1, weight_in_kg 10), (baggage_id 2, weight_in_kg 20), (baggage_id 3, weight_in_kg 30),
(baggage_id 4, weight_in_kg 40), (baggage_id 5, weight_in_kg 50), (baggage_id 6, weight_in_kg 55);
-- Task 12
CREATE INDEX idx12 ON baggage(weight_in_kg);
EXPLAIN ANALYZE
SELECT * FROM baggage WHERE weight_in_kg > 49;
EXPLAIN ANALYZE 368 ms
SELECT * FROM baggage WHERE weight_in_kg > 5;
Files
default ~/DataGripProjects/default
Scratches and Consoles
Services
Database
postgres@localhost
console_3 100 ms
console
console_2 368 ms
console_1
Output
Result 47
Result 46
CSV
QUERY PLAN
Seq Scan on baggage (cost=0.00..1.07 rows=2 width=16) (actual time=0.027..0.029 rows=6.00 loops=1)
Filter: (weight_in_kg > '5'::numeric)
Buffers: shared hit=1
5 rows
Database Consoles > postgres@localhost > console_2 [postgres@localhost]
135:1 LF UTF-8 4 spaces
Trial


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console [postgres@localhost]
console_1 [postgres@localhost]
console_2 [postgres@localhost]
Tx: Auto
Playground
airport_lab6.public
-- Task 12
CREATE INDEX idx12 ON baggage(weight_in_kg);
EXPLAIN ANALYZE
SELECT * FROM baggage WHERE weight_in_kg > 49;
EXPLAIN ANALYZE
SELECT * FROM baggage WHERE weight_in_kg > 5;
-- Task 13
CREATE INDEX IF NOT EXISTS idx13 ON airport(airport_name);
EXPLAIN ANALYZE
SELECT * FROM airport WHERE airport_name LIKE 'A%';
EXPLAIN ANALYZE
SELECT * FROM airport WHERE airport_name LIKE '%Airport'; 355 ms
Files
default ~/DataGripProjects/default
Scratches and Consoles
Services
Database
postgres@localhost
console_3 100 ms
console
console_2 355 ms
console_1
Output
Result 47
airport_lab6.public.airport
Tx: Auto
DDL
CSV
airport_id
airport_name
city
country
1 1 Almaty International Airport Almaty Kazakhstan
2 2 Astana International Airport Astana Kazakhstan
3 3 Istanbul Airport bul Turkey
4 4 Frankfurt Airport Frankfurt Germany
5 rows
Database Consoles > postgres@localhost > console_2 [postgres@localhost]
144:58 LF UTF-8 4 spaces
Trial


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Server Objects
console [postgres@localhost]
console_1 [postgres@localhost]
console_2 [postgres@localhost]
Tx: Auto
Playground
airport_lab6.public
-- Task 14
CREATE INDEX IF NOT EXISTS idx14 ON airport(country, city);
EXPLAIN ANALYZE
SELECT * FROM airport
WHERE country = 'Kazakhstan';
EXPLAIN ANALYZE
SELECT * FROM airport
WHERE city = 'Almaty';
EXPLAIN ANALYZE 368 ms
SELECT * FROM airport
WHERE country = 'Kazakhstan'
AND city = 'Almaty';
Files
default ~/DataGripProjects/default
Scratches and Consoles
Services
postgres@localhost
console_3 100 ms
console
console_2 368 ms
console_1
Output
Result 53
Result 54
CSV
QUERY PLAN
Seq Scan on airport (cost=0.00..1.07 rows=1 width=658) (actual time=0.060..0.061 rows=1.00 loops=1)
Filter: (((country)::text = 'Kazakhstan'::text) AND ((city)::text = 'Almaty'::text))
Rows Removed by Filter: 4
6 rows
Database Consoles > postgres@localhost > console_2 [postgres@localhost]
156:1 LF UTF-8 4 spaces
Trial


default
Version Control
Database Explorer
postgres@localhost 2 of 3
airport_lab6 1 of 3
public
tables 6
Database Objects
virtual views 1
postgres 1 of 3
Server Objects
console [postgres@localhost]
console_1 [postgres@localhost]
console_2 [postgres@localhost]
Tx: Auto
Playground
airport_lab6.public
EXPLAIN ANALYZE
SELECT * FROM airport
WHERE country = 'Kazakhstan'
AND city = 'Almaty';
-- Task 15
CREATE INDEX IF NOT EXISTS idx15
ON passenger(LOWER(last_name));
EXPLAIN ANALYZE
SELECT * FROM passenger
WHERE LOWER(last_name) = 'irnis';
EXPLAIN ANALYZE 357 ms
SELECT * FROM passenger
WHERE last_name = 'Irnis';
Files
default ~/DataGripProjects/default
Scratches and Consoles
Services
postgres@localhost
console_3 100 ms
console
console_2 357 ms
console_1
Output
airport_lab6.public.passenger
Result 57
CSV
QUERY PLAN
Seq Scan on passenger (cost=0.00..1.10 rows=1 width=680) (actual time=0.150..0.151 rows=0.00 loops…
Filter: ((last_name)::text = 'Irnis'::text)
Rows Removed by Filter: 8
6 rows
Database Consoles > postgres@localhost > console_2 [postgres@localhost]
168:2 LF UTF-8 4 spaces
Trial


default
Version Control
Database Explorer
postgres@localhost 2 of 3
airport_lab6 1 of 3
public
tables 6
Database Objects
virtual views 1
postgres 1 of 3
Server Objects
console [postgres@localhost]
console_1 [postgres@localhost]
console_2 [postgres@localhost]
Tx: Auto
Playground
airport_lab6.public
EXPLAIN ANALYZE
SELECT * FROM passenger
WHERE LOWER(last_name) = 'irnis';
EXPLAIN ANALYZE
SELECT * FROM passenger
WHERE last_name = 'Irnis';
-- Task 16
CREATE INDEX IF NOT EXISTS idx16
ON booking(passenger_id)
INCLUDE (ticket_price);
EXPLAIN ANALYZE
SELECT ticket_price 351 ms
FROM booking
WHERE passenger_id = 1;
Files
default ~/DataGripProjects/default
Scratches and Consoles
Services
Database
postgres@localhost
console_3 100 ms
console
console_2 351 ms
console_1
Output
airport_lab6.public.booking
Result 57
DDL
CSV
ticket_price
1 45000.00
2 180000.00
2 rows
Database Consoles > postgres@localhost > console_2 [postgres@localhost]
179:24 LF UTF-8 4 spaces
Trial


default
Version Control
Database Explorer
postgres@localhost 2 of 3
airport_lab6 1 of 3
public
tables 6
Database Objects
virtual views 1
postgres 1 of 3
Server Objects
console [postgres@localhost]
console_1 [postgres@localhost]
console_2 [postgres@localhost]
Tx: Auto
Playground
airport_lab6.public
WHERE passenger_id = 1;
-- Task 17
ALTER TABLE airline
ADD COLUMN IF NOT EXISTS airline_code VARCHAR(10);
CREATE UNIQUE INDEX IF NOT EXISTS idx17
ON airline(airline_code);
SELECT indexname 360 ms
FROM pg_indexes
WHERE tablename = 'airline';
Files
default ~/DataGripProjects/default
Scratches and Consoles
Services
postgres@localhost
console_3 100 ms
console
console_2 360 ms
console_1
Output
airport_lab6.public.booking
airport_lab6.pg_catalog.pg_indexes
DDL
CSV
indexname
1 airline_pkey
2 idx17
2 rows
Database Consoles > postgres@localhost > console_2 [postgres@localhost]
190:29 LF UTF-8 4 spaces
Trial